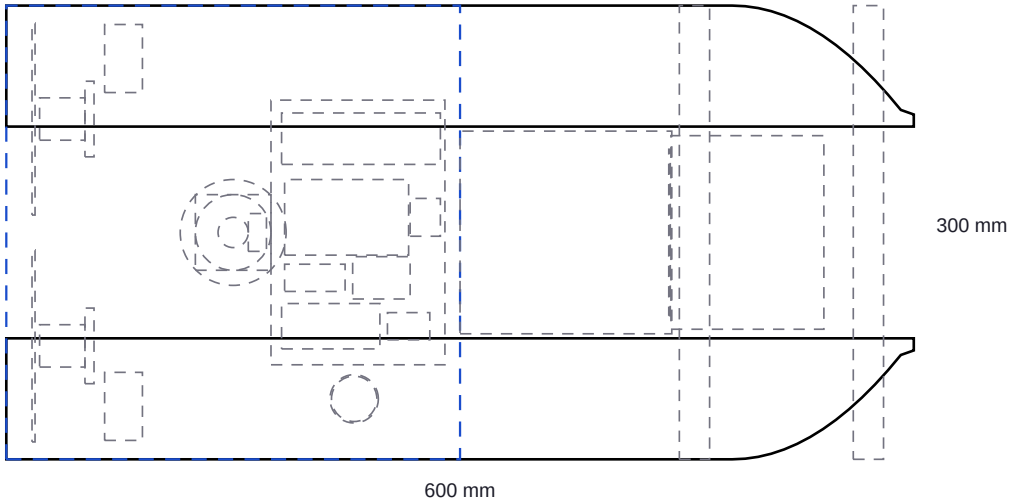
<- must measure 50 mm

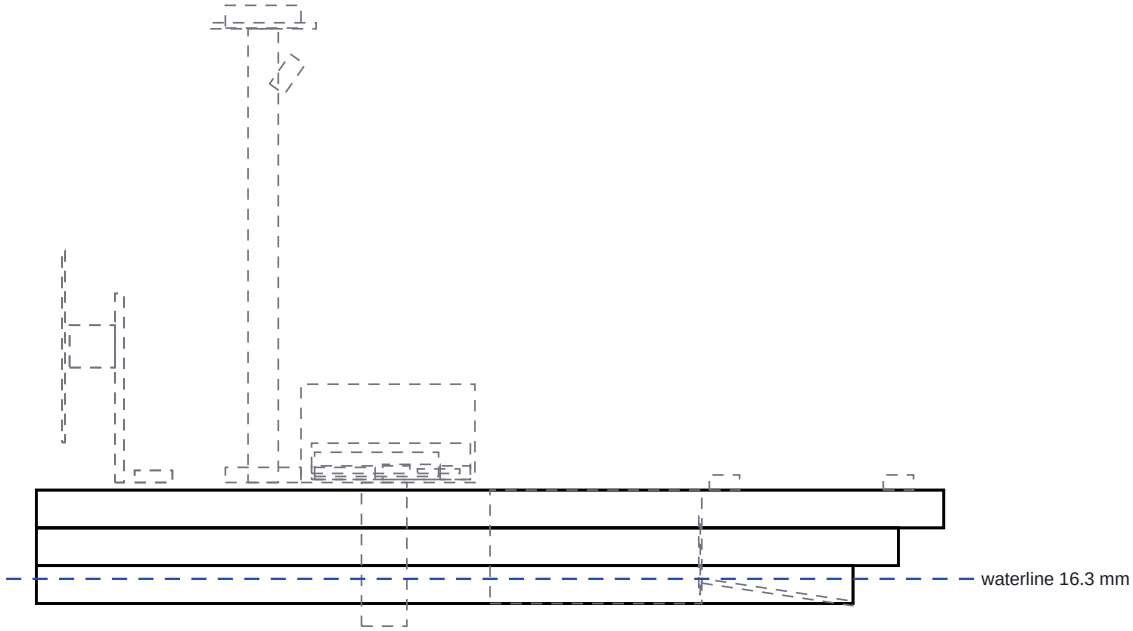
# VARUNA Twin-Hull Skimmer

600 x 300 mm thermocol boat | Recommended

**PLAN (from above), scale 1:5**




**SIDE, scale 1:5**




| | |
|---|---|
| **Size** | 600 x 300 mm, hull 75 mm tall, 396 mm to GPS top |
| **Weight** | 1289 g empty, 1558 g with 269 g of plastic |
| **Draft** | 16.3 mm empty, 19.7 mm full |
| **Reserve buoyancy** | 4.7 kg before the deck goes under |
| **Stability (GM)** | 671 mm: very stable |
| **Trim** | bow 14.9 mm higher than stern empty; slide the battery to level |
| **Collection** | 1.34 L basket |
| **Camera** | 334 mm above water, tilted 35 deg down |
| **Battery** | LiPo 3S 2200 mAh (24 Wh), about 0.8 h |
| **Thermocol** | 1 sheet 1000 x 500 x 25 mm |

## How to use these templates

1. Print at 100% / Actual size. Check the 50 mm bar with a ruler.
2. Join tiles by matching the grey 50 mm grid and the tile letters.
3. Stick the paper on thermocol with a glue stick.
4. Cut with a hot-wire cutter or a new snap-off blade, held upright.
5. Stack layers on skewers through the + holes. Glue with Fevicol SH.
6. Never use Fevikwik, spray paint or solvent glue: they melt thermocol.
7. Sand the bow steps smooth, coat with PVA + tissue or packing tape, then water-based acrylic paint.
8. Float-test in a tub before fitting the electronics.

cut line

cut out (window / pocket)

fold / glue line

placement guide, do not cut

skewer hole: lines up the layers

<- must measure 50 mm

A1

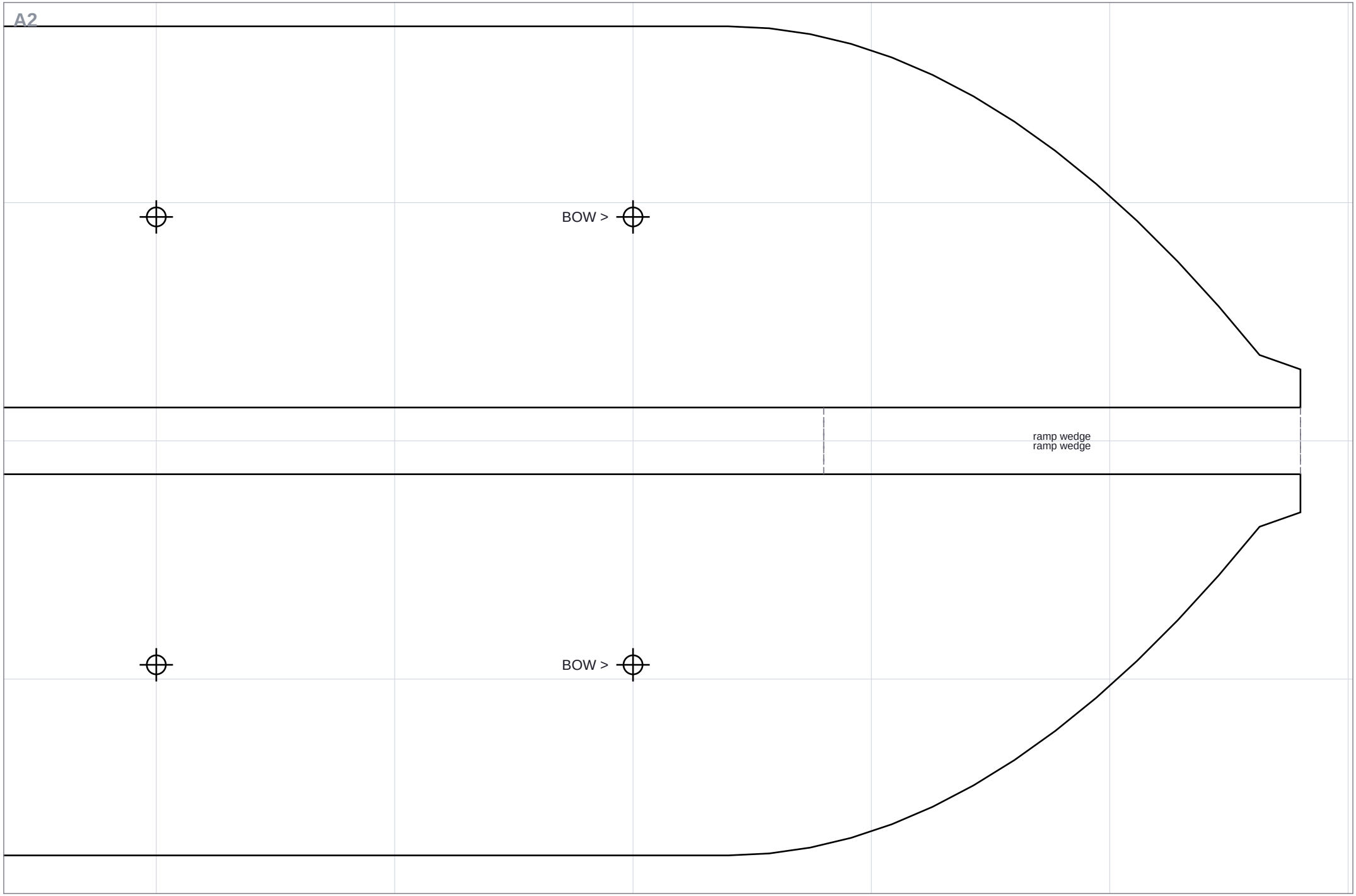
LEFT pontoon - layer 1 (bottom)

overlap: tile A2 goes on top

probe hole 32 mm

RIGHT pontoon - layer 1 (bottom)

<- must measure 50 mm



<- must measure 50 mm

A1

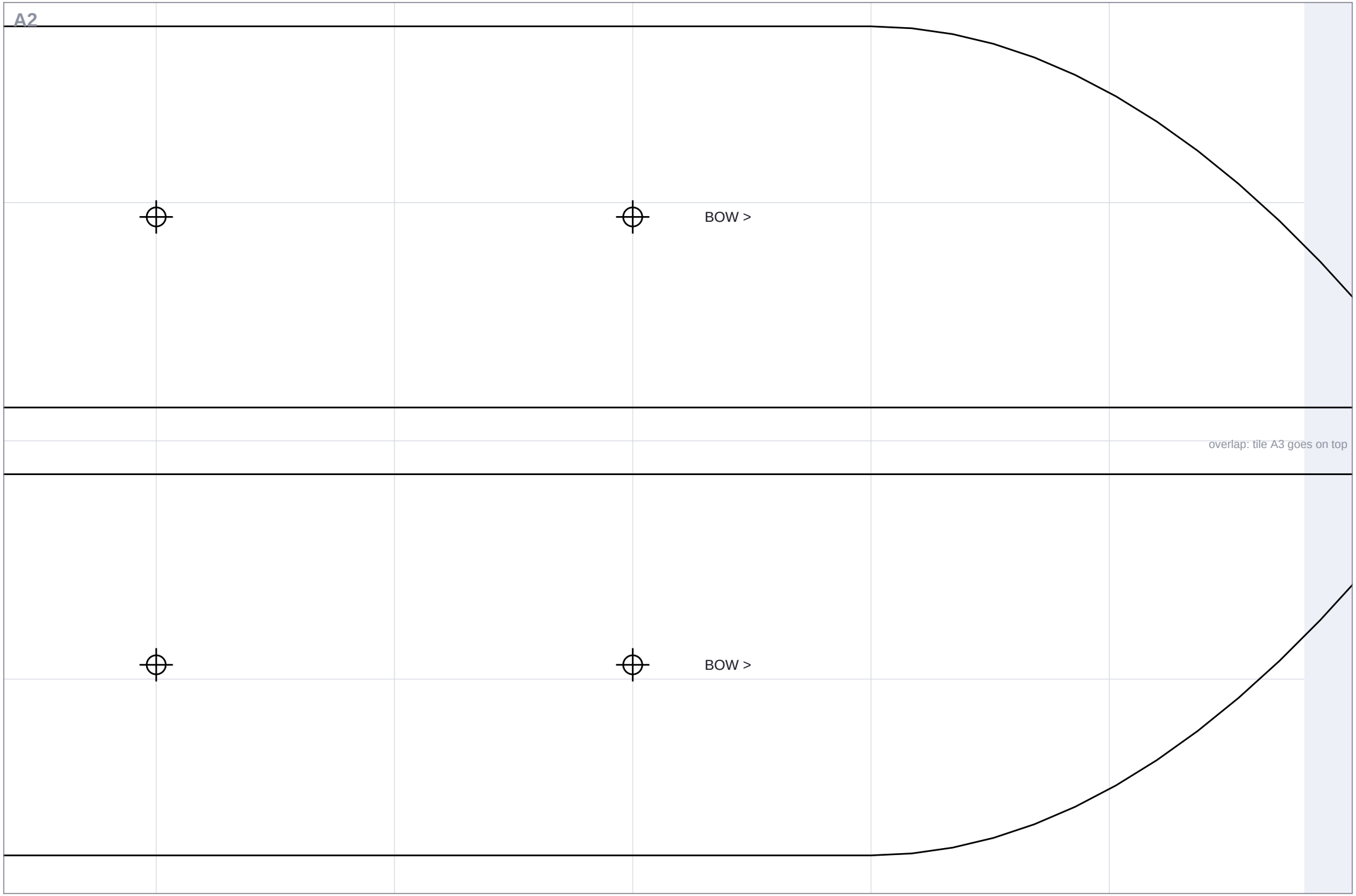
LEFT pontoon - layer 2 (middle)

overlap: tile A2 goes on top

probe hole 32 mm

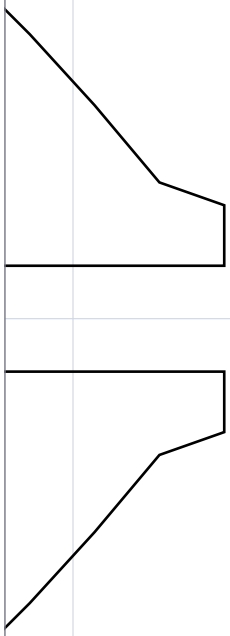
RIGHT pontoon - layer 2 (middle)

<- must measure 50 mm



<- must measure 50 mm

A3



<- must measure 50 mm

A1

LEFT pontoon - layer 3 (top)

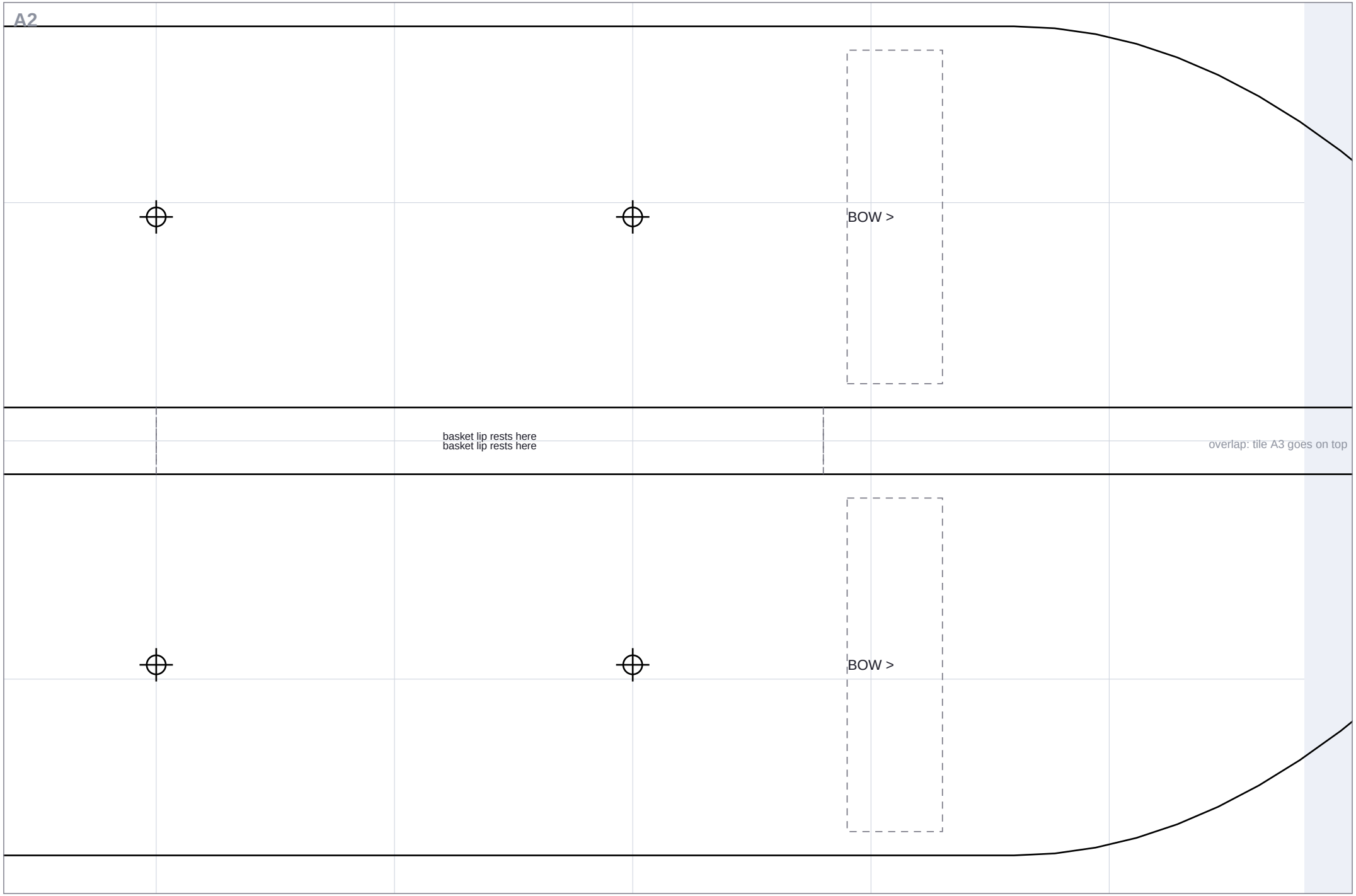
overlap: tile A2 goes on top

probe hole 32 mm

RIGHT pontoon - layer 3 (top)

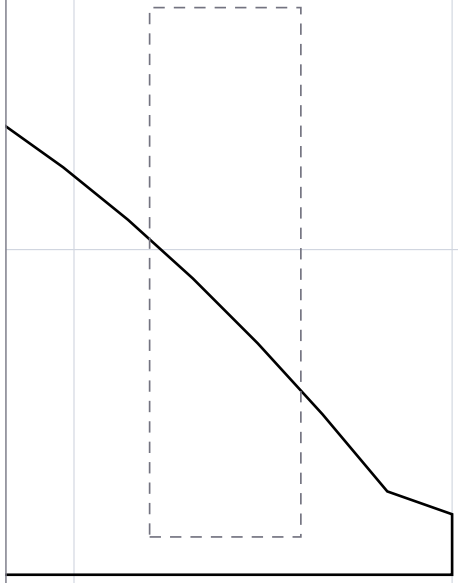
# VARUNA Twin-Hull Skimmer | Pontoons layer 3 (top), left + right - tile A2 of A3 (25 mm thermocol)

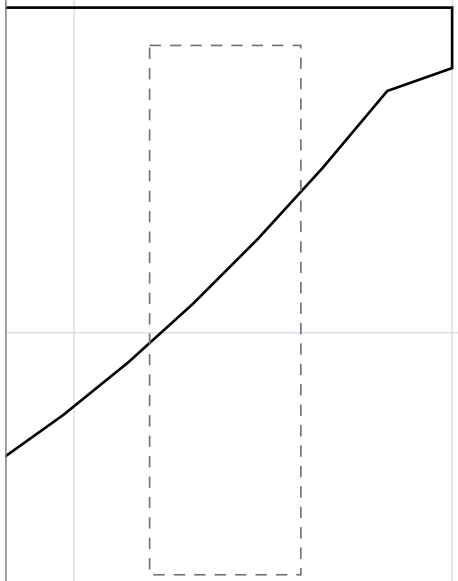
<- must measure 50 mm



<- must measure 50 mm

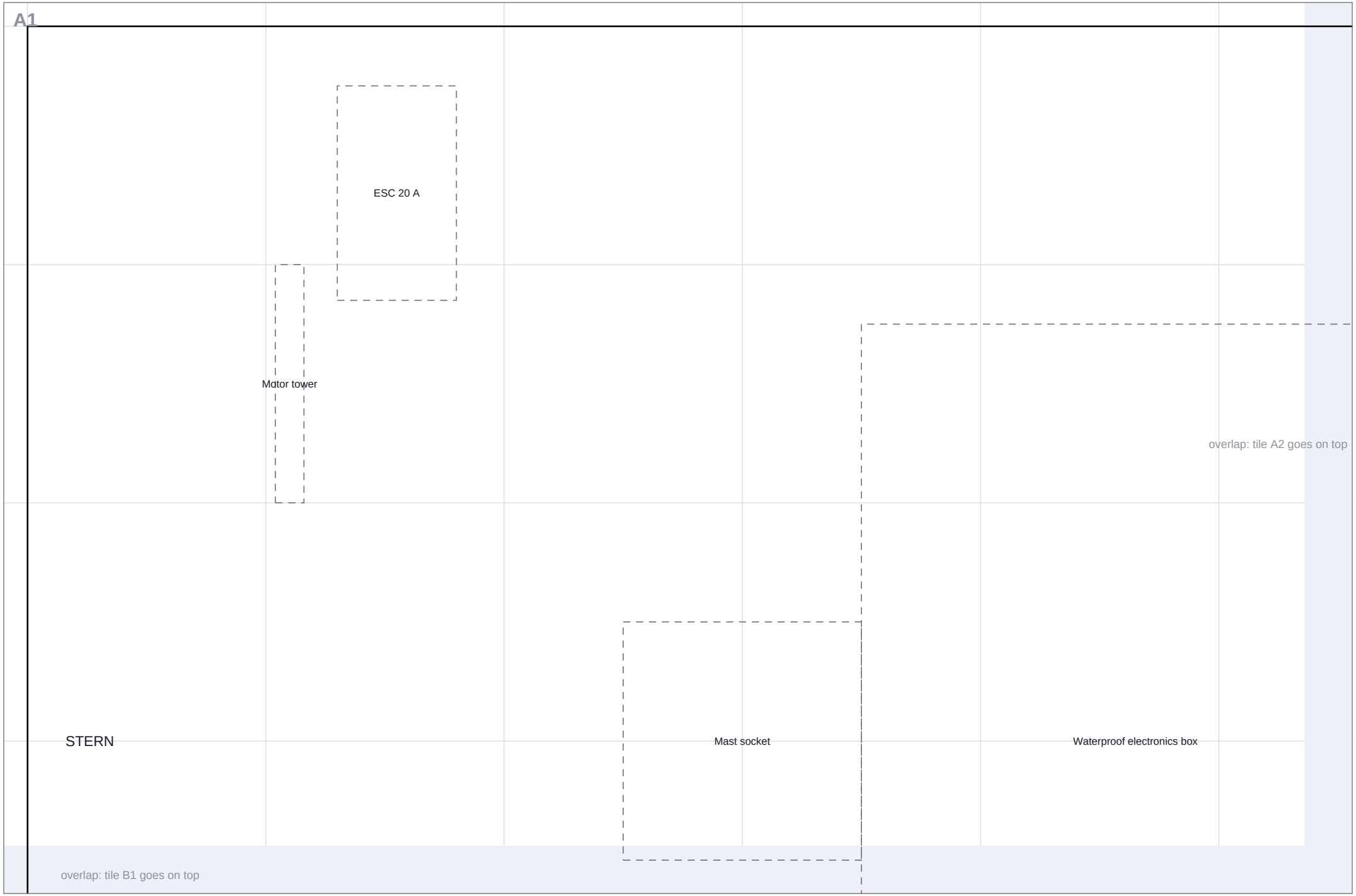
A3





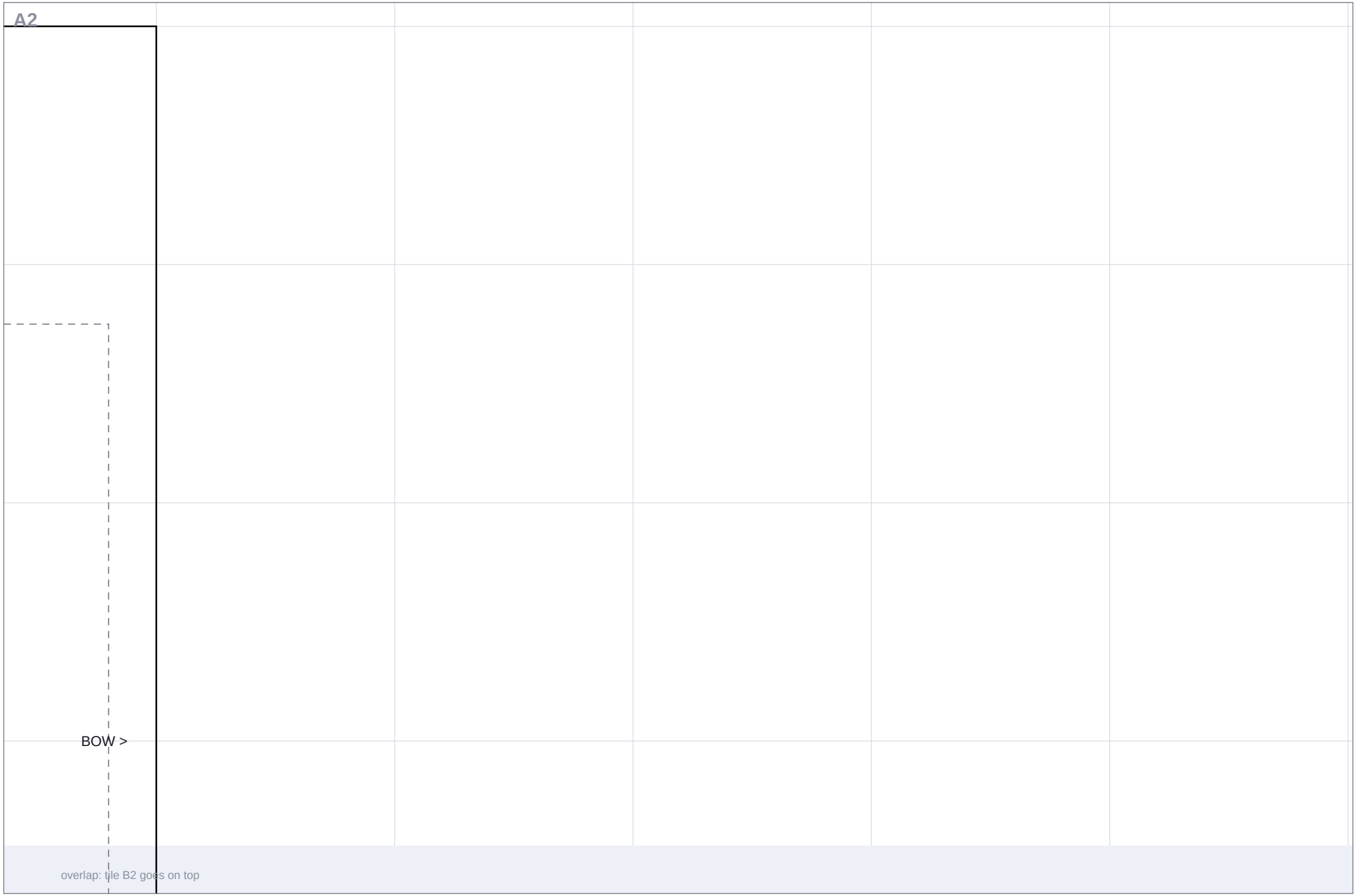
# VARUNA Twin-Hull Skimmer | Deck - tile A1 of B2 (5 mm sunboard)

<- must measure 50 mm



# VARUNA Twin-Hull Skimmer | Deck - tile A2 of B2 (5 mm sunboard)

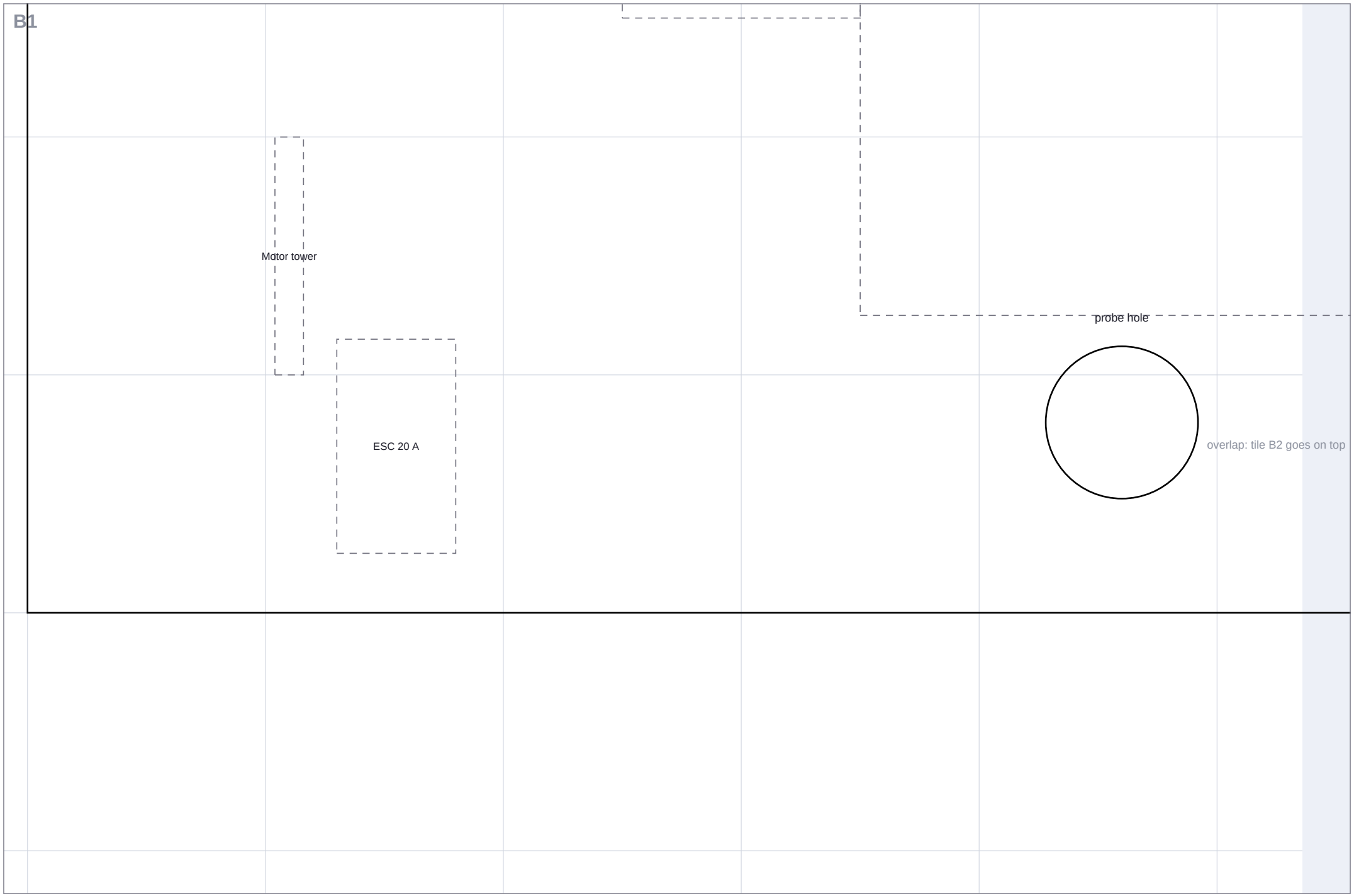
<- must measure 50 mm



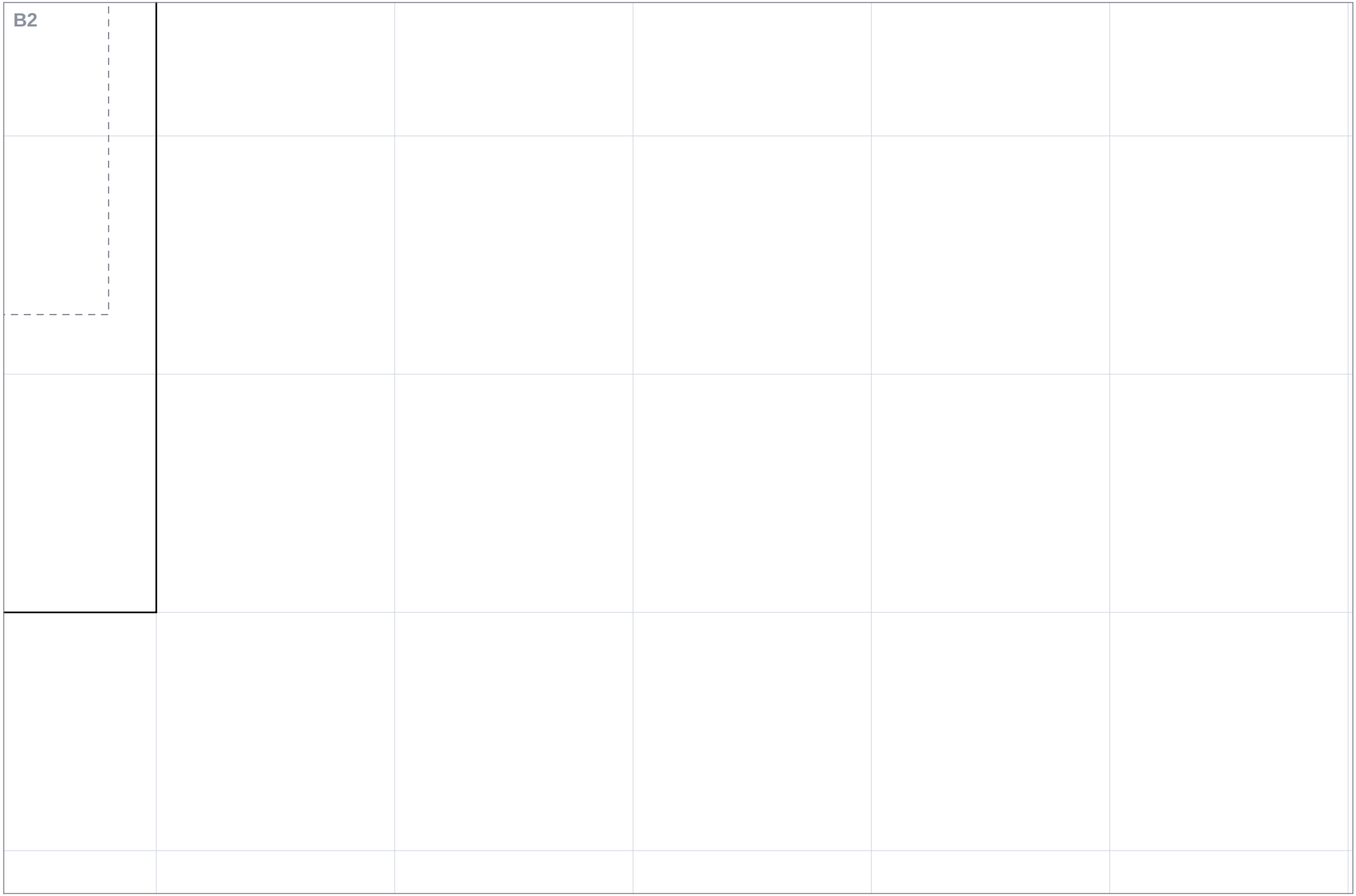
# VARUNA Twin-Hull Skimmer | Deck - tile B1 of B2 (5 mm sunboard)



<- must measure 50 mm

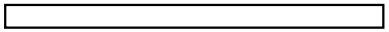


<- must measure 50 mm



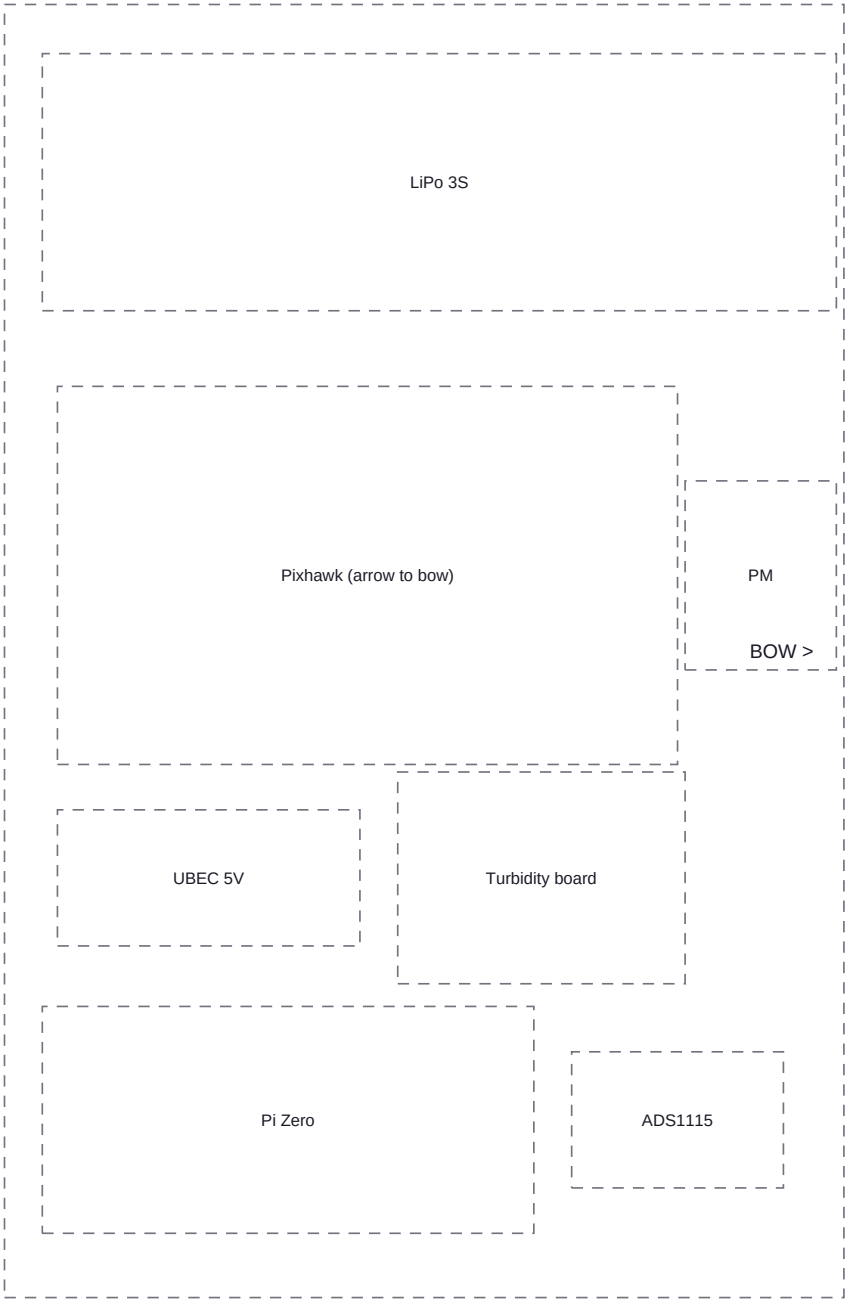
# VARUNA Twin-Hull Skimmer | small parts 1 of 7



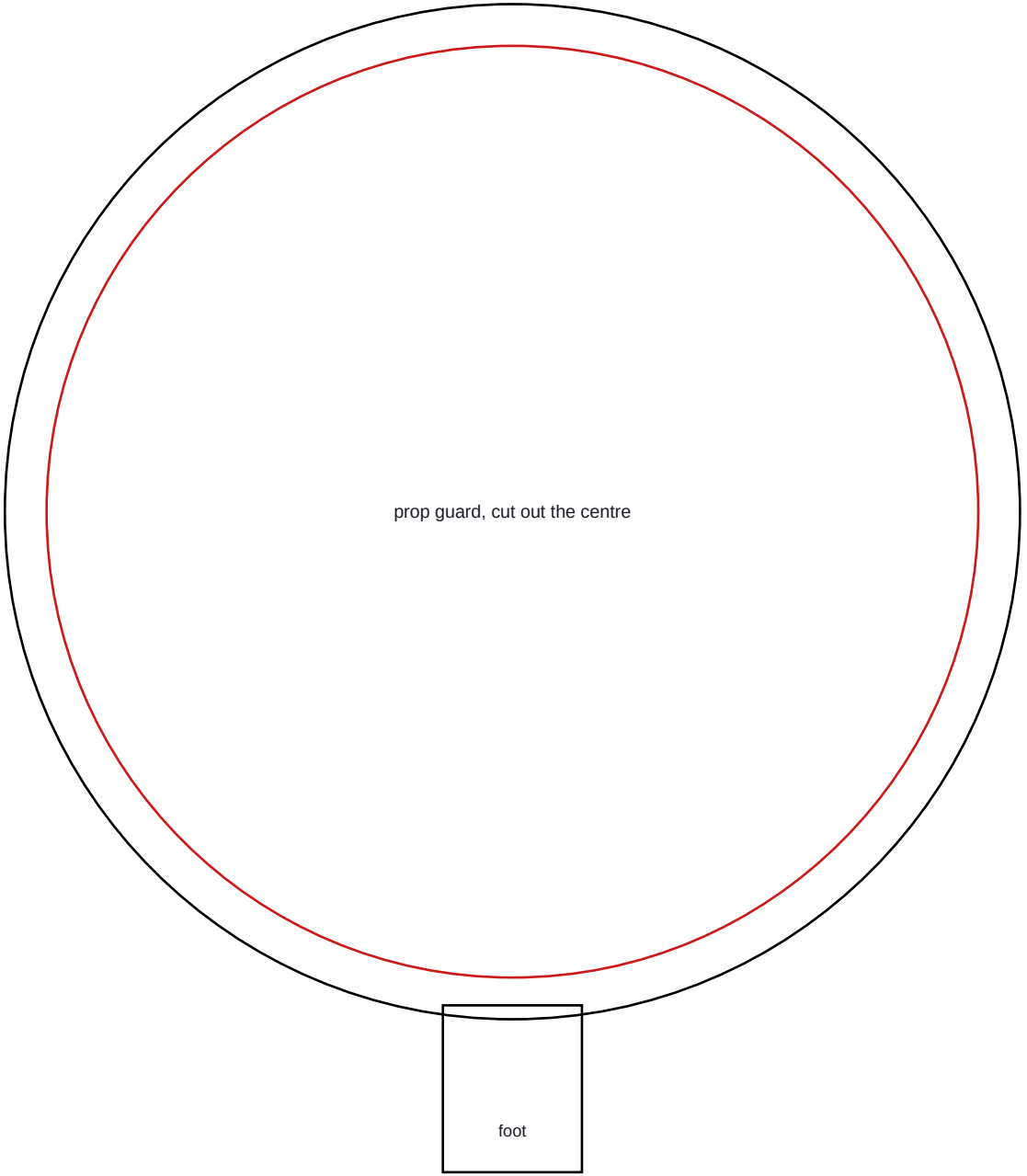


<- must measure 50 mm

**Electronics box floor guide**
paper guide, glue inside the floor

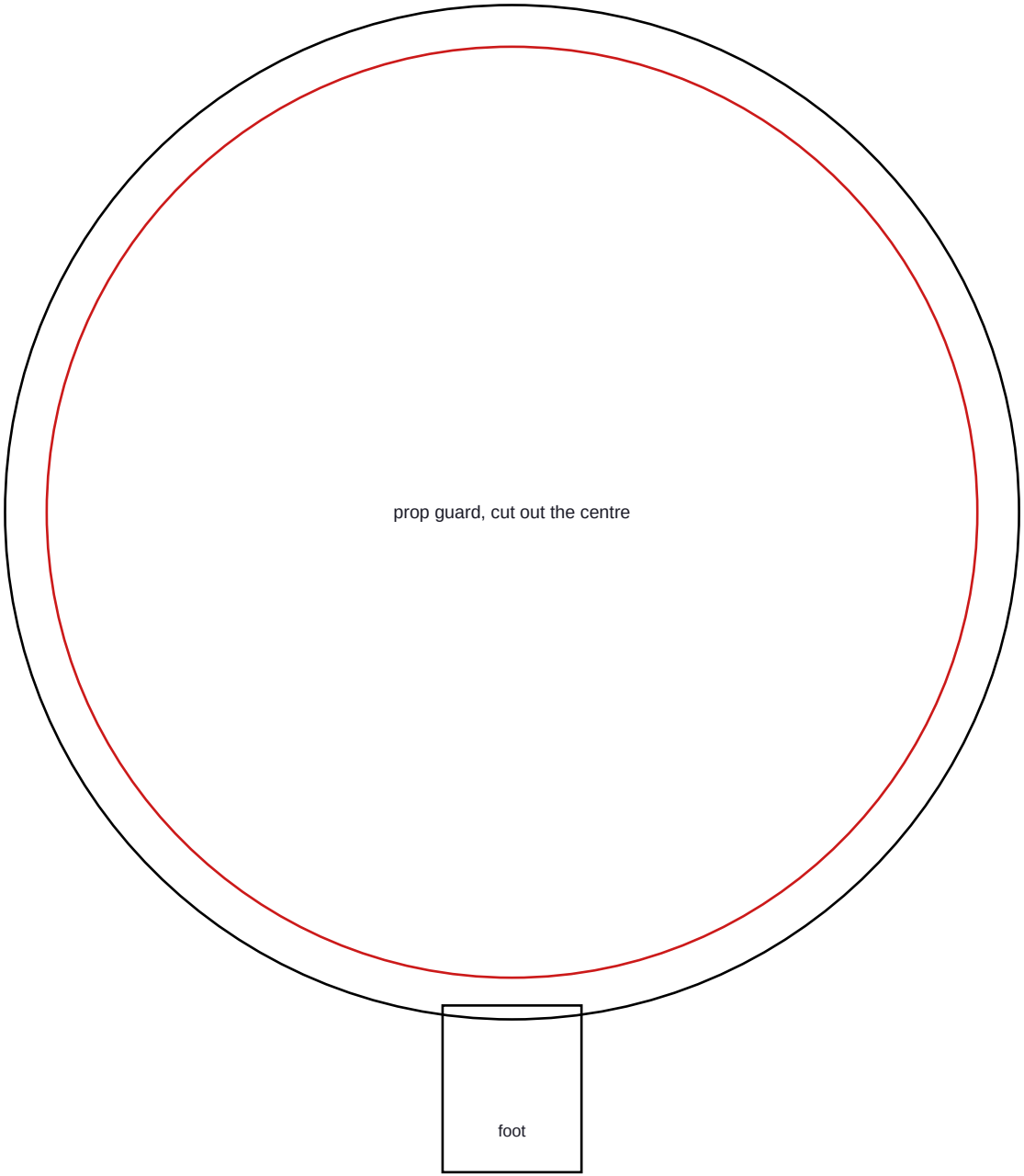



**Prop guard ring (left)**
5 mm sunboard



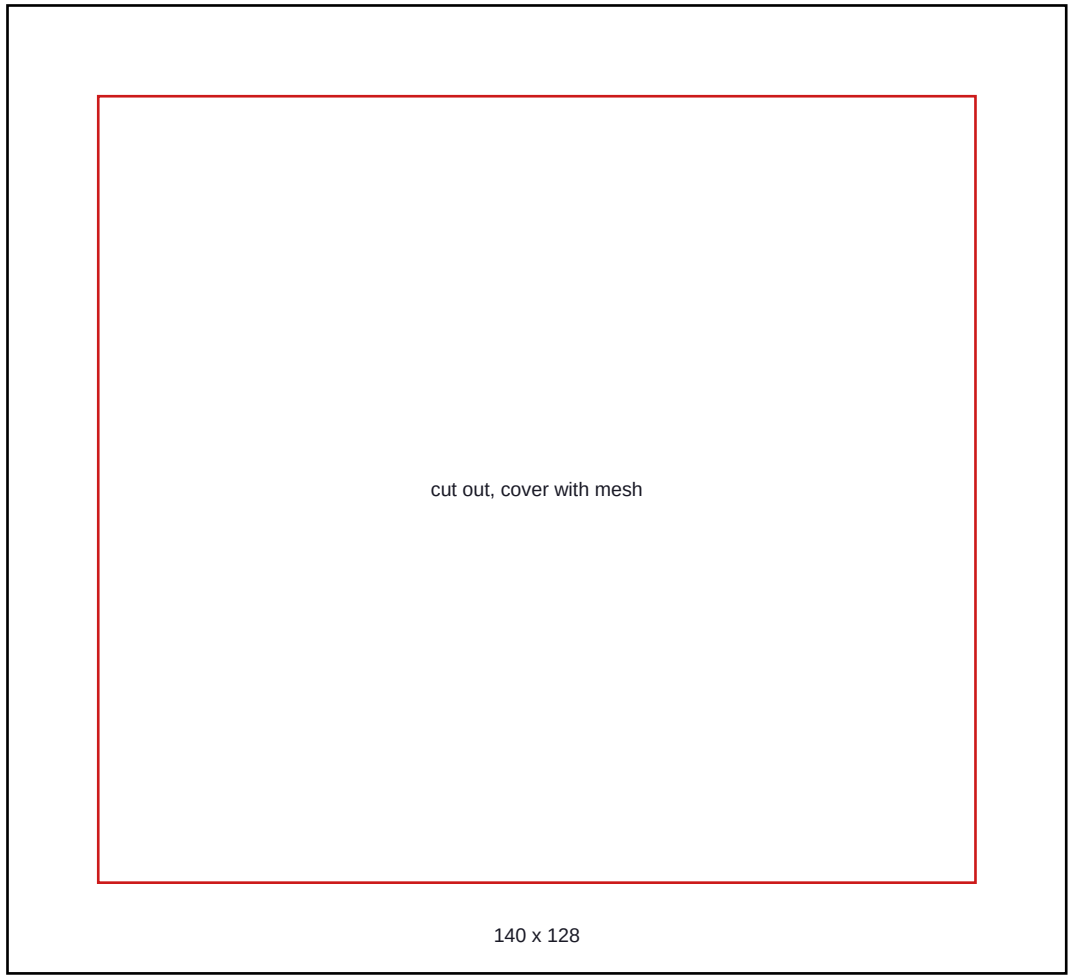
<- must measure 50 mm

**Prop guard ring (right)**
5 mm sunboard

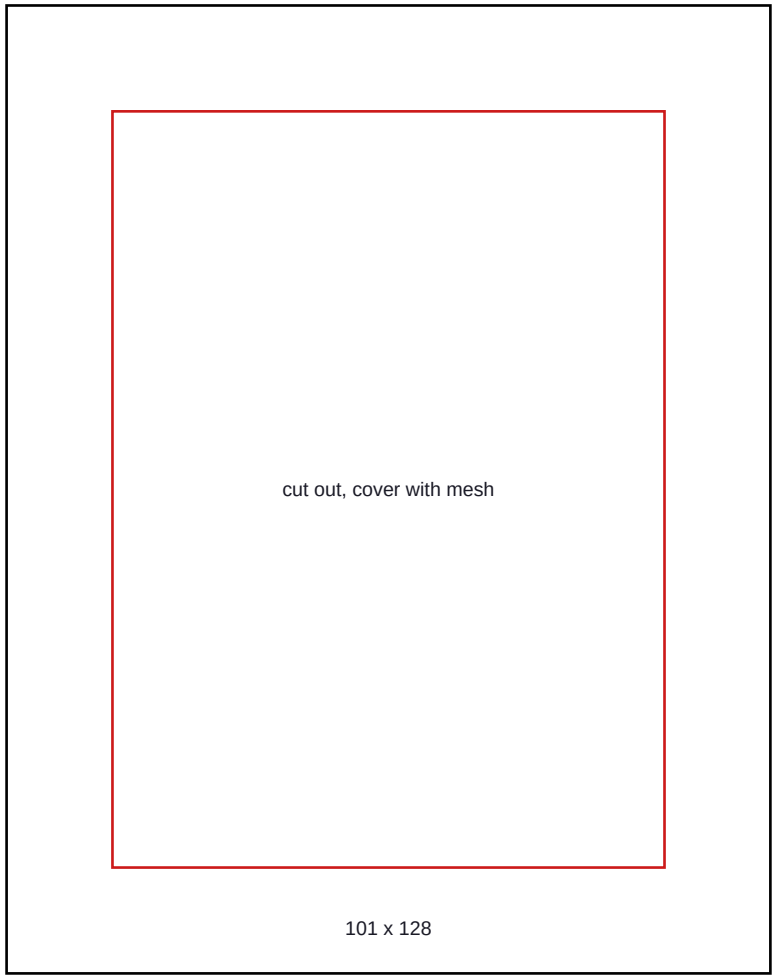


<- must measure 50 mm

**Basket bottom**
3 mm sunboard

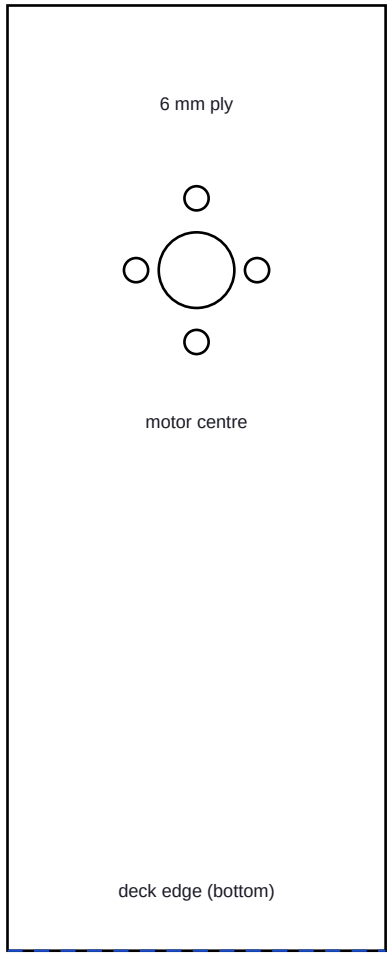



**Intake ramp frame**
3 mm sunboard

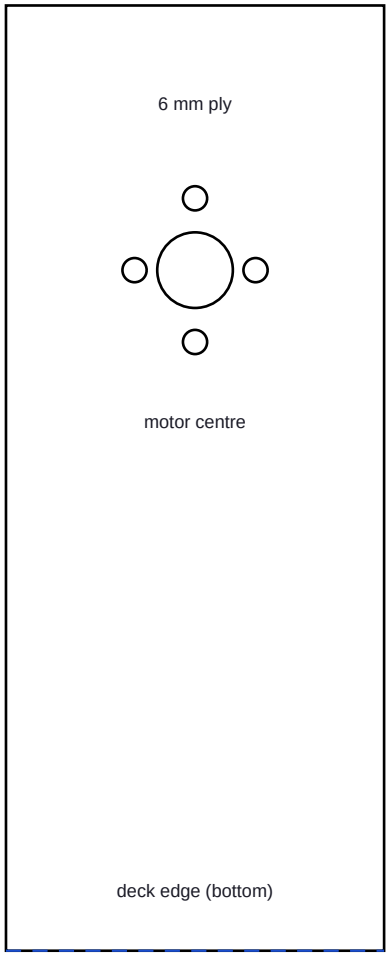


<- must measure 50 mm

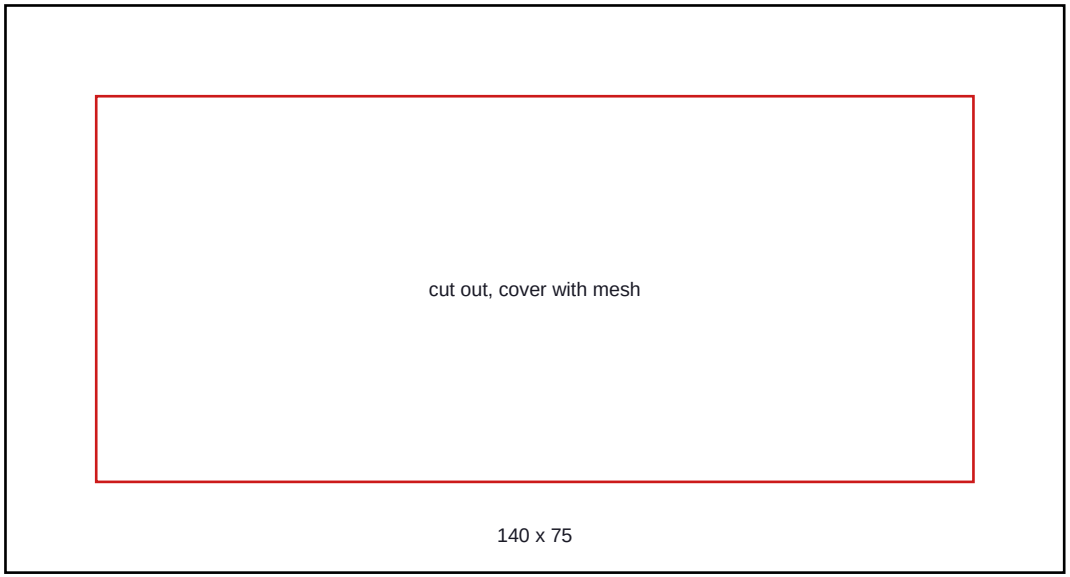
**Motor tower plate (left)**
6 mm plywood (or 2 x 5 mm sunboard)




**Motor tower plate (right)**
6 mm plywood (or 2 x 5 mm sunboard)

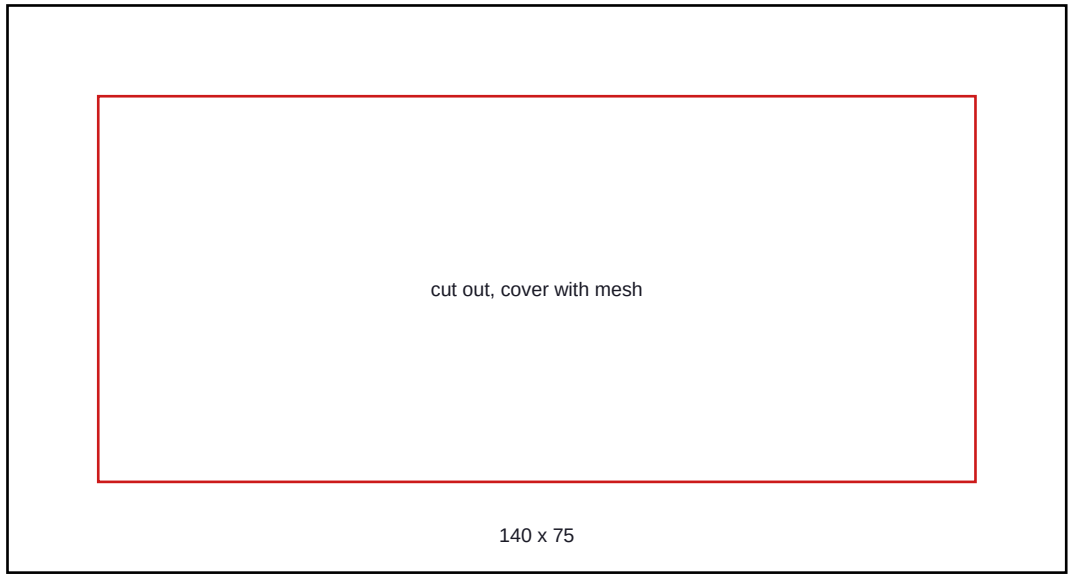



**Basket side (left)**
3 mm sunboard

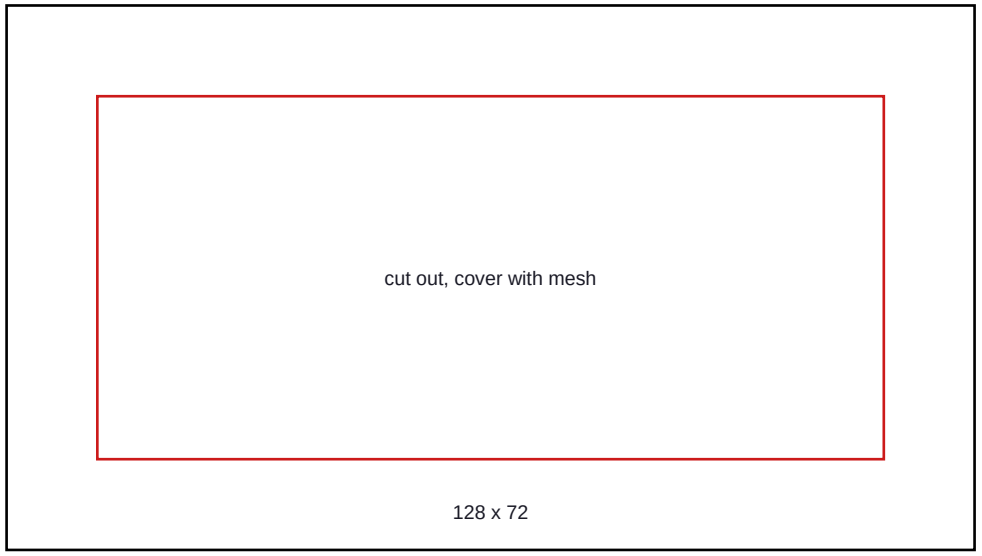


<- must measure 50 mm

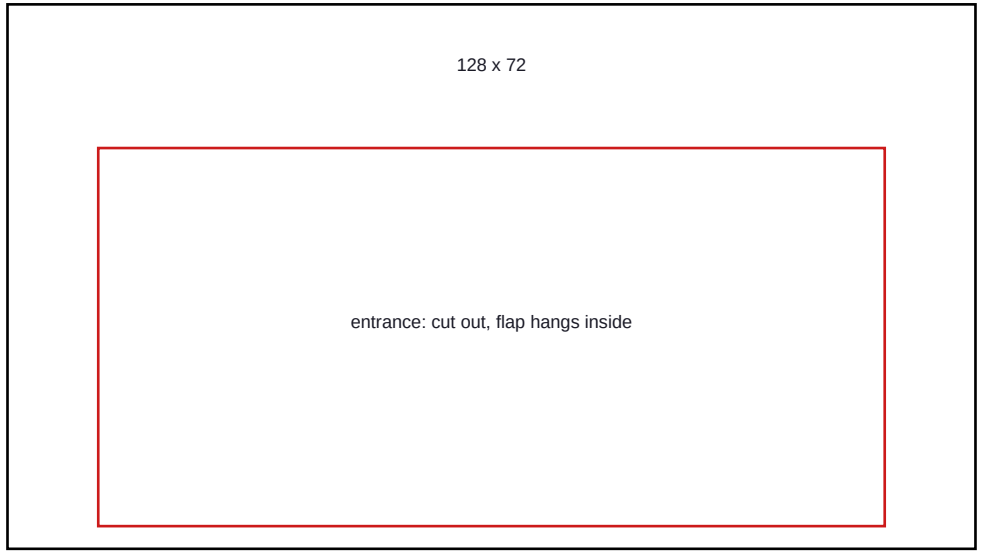
**Basket side (right)**
3 mm sunboard



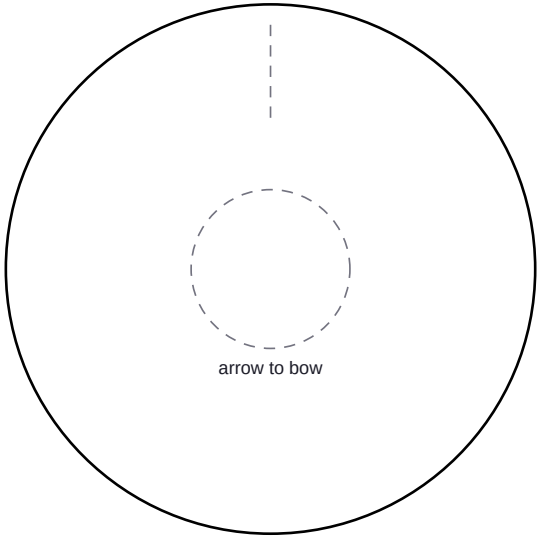

**Basket back end**
3 mm sunboard



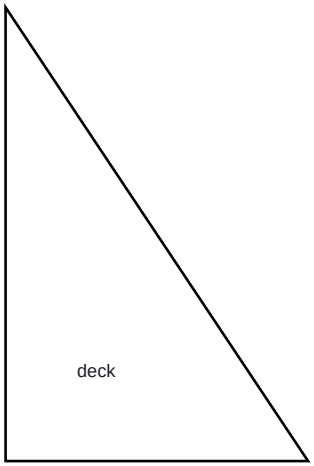

**Basket front end (entrance)**
3 mm sunboard




**GPS disc**
5 mm sunboard

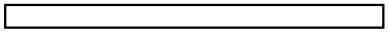



**Tower gusset**
5 mm sunboard



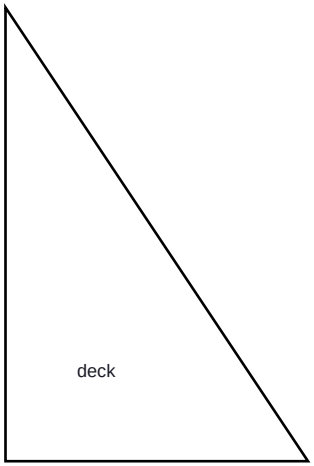
# VARUNA Twin-Hull Skimmer | small parts 6 of 7



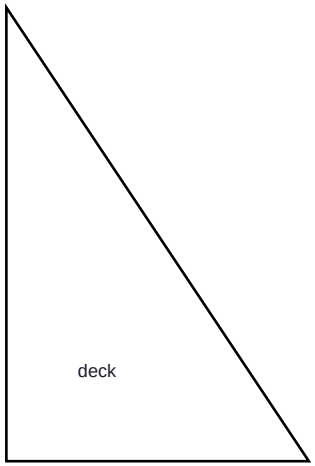


<- must measure 50 mm

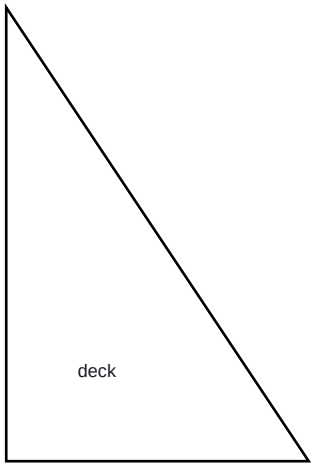
**Tower gusset**
5 mm sunboard



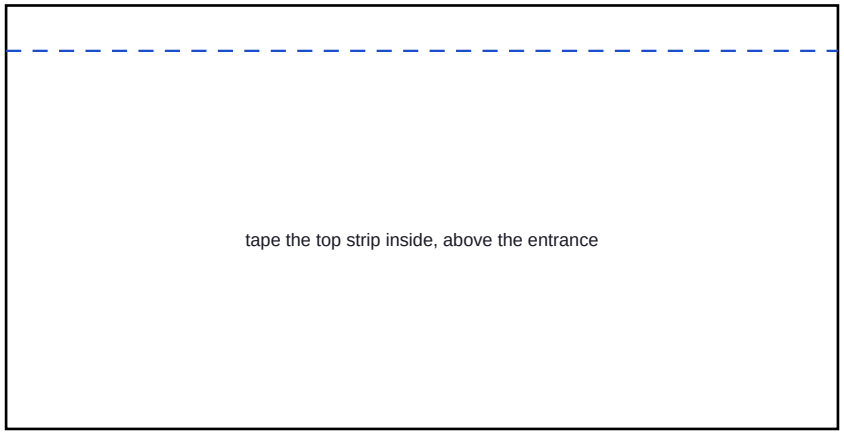

**Tower gusset**
5 mm sunboard



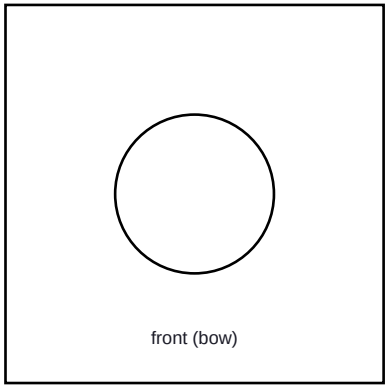

**Tower gusset**
5 mm sunboard



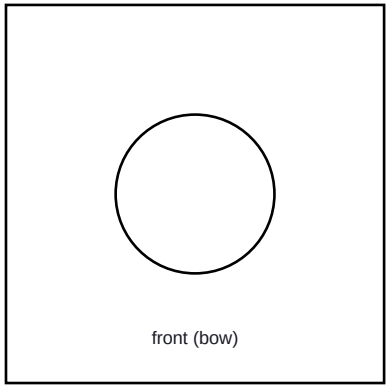

**One-way flap**
PET sheet from a bottle / transparency



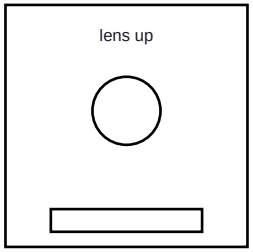

**Mast socket layer**
5 mm sunboard (cut 2, top one with hole)



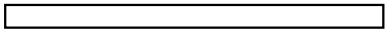

**Mast socket layer**
5 mm sunboard (cut 2, top one with hole)




**Camera plate**
3 mm sunboard



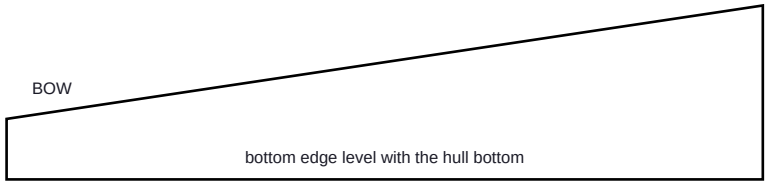

**Basket lip (left)**
3 mm sunboard



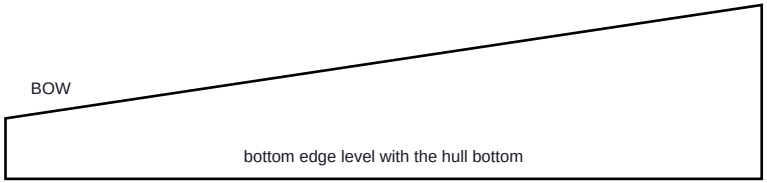
<- must measure 50 mm

**Basket lip (right)**
3 mm sunboard

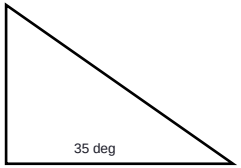
**Ramp support wedge**
5 mm sunboard (cut 2)

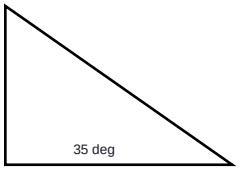



**Ramp support wedge**
5 mm sunboard (cut 2)




**Camera wedge side (35 deg)**
3 mm sunboard




**Camera wedge side (35 deg)**
3 mm sunboard